

Algorithms Final

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Description

Our program is a visual representation of a living room. There are multiple things/objects that you can interact with. Firstly, the dog in the corner can be pet, when holding the left mouse button and moving over him, hairs will fly around (these hairs are particles). The next interaction in our program is the hand and the ball. When pulling the hand back and releasing it, you can throw the ball. If the ball is thrown against the window, it will crack; if the ball hits the window a second time, it will fully break. Pressing the 'R' button will reset the position of the ball. Next, the mass spring damper system is used in the water bowl next to the dog, clicking on the bowl will create splashes in the water. Clicking on the television will display static, which is generated by Perlin noise. When starting the program, a painting is generated next to the window. The pattern in this painting is determined by the normal distribution. Behind the window, a flock of birds flies around which logic is handled with Shiffman flocking. This flock is affected by the ball as the birds avoid it when the ball flies past them. When clicking the painting, the normal distribution gets reset. When the hairs from the hair system fly into the water bowl, they create a splash. When the window is broken, a wind force also acts on the hairs. When these hairs fall on the ground, they linger for a set amount of time before they get removed.

Class diagram

